

Placements Submission Approval Form

2024-25 DESIGN ENGINEERING INDUSTRIAL PLACEMENTS

Form for employers to approve coursework submissions for assessment of industrial placement students. Please note that any documents submitted to Imperial College will only be used for the purposes of assessment, and will not be seen by anyone who does not form a part of the assessment team unless otherwise stated.

SECTION 1. DETAILS

Student Name	Imogen Young
Company Name	Imperial College London
Name and position of company representative completing this form.	Rebecca Stewart

SECTION 2. SUBMISSION TYPE

Please tick which submission you are approving on this occasion.

1.	Interim Report: <i>This is a report that will just be used internally for marking the student and tracking their progress.</i>
2.	Final Report: X <i>This is a report that will just be used internally for marking the student.</i>
3.	Communication Piece: <i>This is an online 'info page' in the Placements website, that enable the student to share their placement experiences with a wider audience. The audience for this Exhibition will include Imperial staff and students, as well as representatives from external companies including other current and prospective placement providers.</i>

SECTION 3. APPROVAL

I, the undersigned, declare that:

- I have looked through the submission outlined above.
- It is representative of the student's contributions and work.
- It is approved to be submitted for assessment and does not contain any materials that cannot be seen by the assessment team (or the general public if it is the exhibition).

Signature: _____ Rebecca Stewart _____ Date: _____ 27 September 2025 _____

Please add Company/Organisation stamp if available

If you have any queries about confidentiality or the assessment process, then please contact our Placements team at placements-designeng@imperial.ac.uk



IMPERIAL

IP2 Final Placement Report

Student: Imogen Young 02035025

Role: Research Assistant in the e-Body Lab

April 2025 – September 2025

Word count (excluding covers, contents, captions, and references): 1994

Table of Contents

1 – PLACEMENT SUMMARY	2
1.1 – My role	2
1.2 – Responsibilities and impact.....	2
1.2.1 – Outreach events	2
1.2.2 – Academic research	3
1.2.3 – Website rebranding	5
1.3 – Prior expectations.....	6
2 – PROFESSIONAL DEVELOPMENT	7
2.1 – Use of Design Engineering knowledge.....	7
2.2 – Application of theoretical and practical methods.....	7
2.3 – Provision of technical and commercial leadership.....	7
2.4 – Effective interpersonal skills	8
2.5 – Personal commitment to professional standards.....	8
3 – FUTURE CAREER REFLECTION	9
3.1 – Personal growth	9
3.2 Professional growth	9
3.3 – Strengths, weaknesses, and discoveries	9
3.4 – Further industry opportuntites.....	10
References:	10

1 – PLACEMENT SUMMARY

Imperial is at the forefront of technological research, regularly producing more ‘world-leading’ research than any other UK university [1]. The e-Body Lab research group [2], led by Dr Rebecca Stewart, moved to the Dyson School of Design Engineering in 2019. The lab explores how computing and sensing technologies can be embedded within textiles, publishing in papers such as TEI, DIS, and IEEE.

1.1 – My role

As a research assistant, I contributed to research projects, publications, and outreach activities to promote the Lab and secure future funding. I gained skills in academic research and science communication specifically, alongside growing my experience in prototyping and coding.

1.2 – Responsibilities and impact

1.2.1 – Outreach events

I worked with a D&T Teacher at the Marlborough Primary School to develop a workshop introducing year 5 students to sewing electrical circuits [Figure 1]. I also created an interactive activity to engage visitors in spatial audio research at the Great Exhibition Road Festival. This improved the visibility of the lab within our community and inspires future generations to continue research in this field.

Achievements

- › Developed three class worksheets of varying levels [Figure 2]. Each has clear instructions and photos, and points of reflection built in to reinforce engineering lessons. Produced an additional delivery guide for teachers, outlining a suggested lesson plan and equipment list. These are now publicly available from the e-Body website [3].
- › Delivered a 2-hour workshop to ~50 students. Supported teachers learn e-textile principles to answer all questions.
- › isolate the ears from existing 3D scans of human heads and created a pair matching game [Figure 3]. Demonstrated why we all perceive sound differently due to unique outer ears.

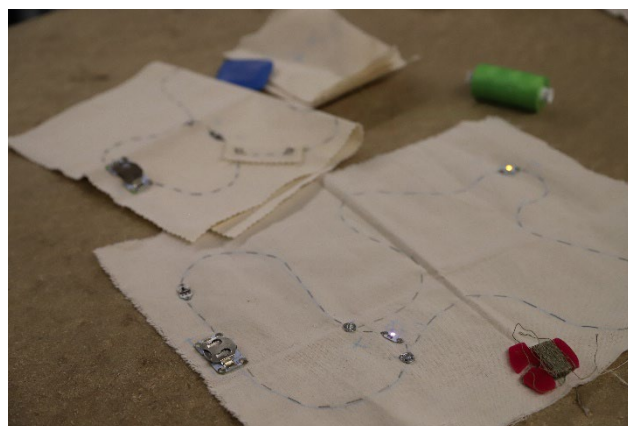


Figure 1 - Example of a sewn LED circuit

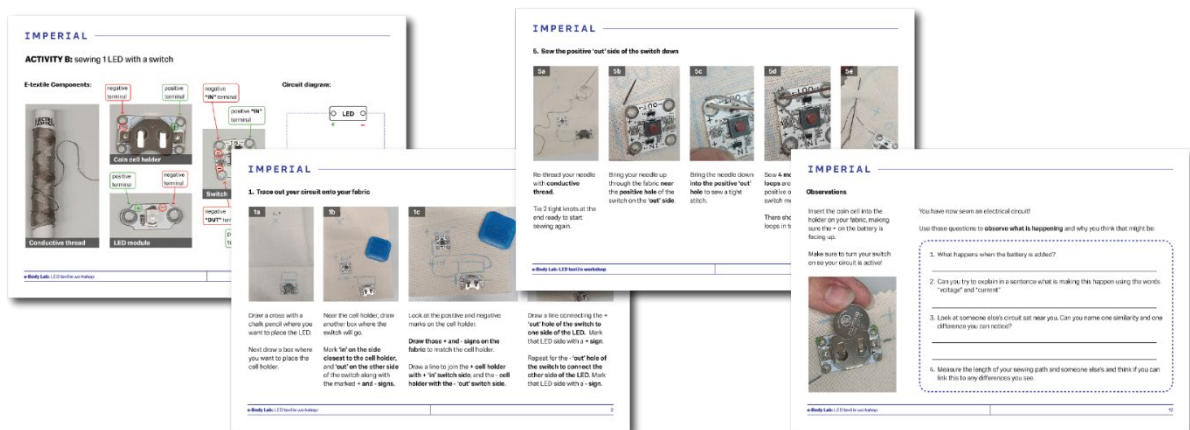


Figure 2 - Sample of the activity worksheets



Figure 3 - Ear pair game at the Great Exhibition Road Festival

1.2.2 – Academic research

I supported a PhD Candidate (Maya), researching “Self-Powered Textile-based Biomonitoring Systems”.

Achievements

- › Designed and 3D printed a new yarn collection device [Figure 4] with variable spindle speeds for controlling coating thickness.
- › Adapted a 3D printer with laser-cut and 3D printed to integrate a load cell and accurately deliver known forces to materials [Figures 5, 6, 7]. Wrote a Python program to characterise triboelectric properties and output data to CSV [Figure 8]. The rig will gather data for an upcoming paper and standardise material characterisation within the lab.

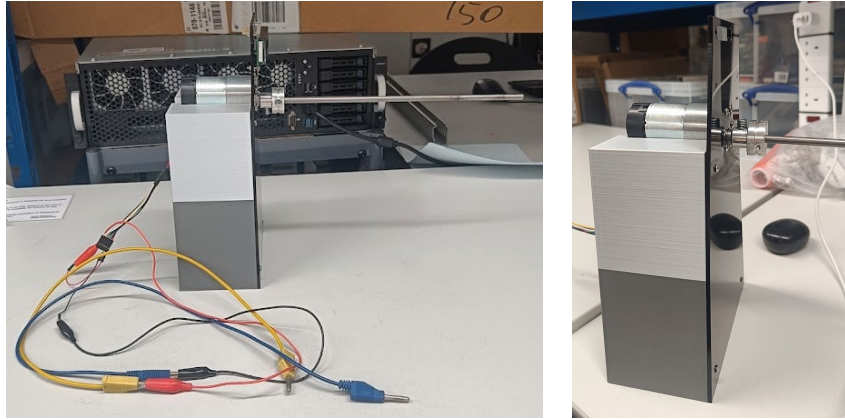


Figure 2 - Yarn collection device



Figure 5 - Triboelectric testing set up

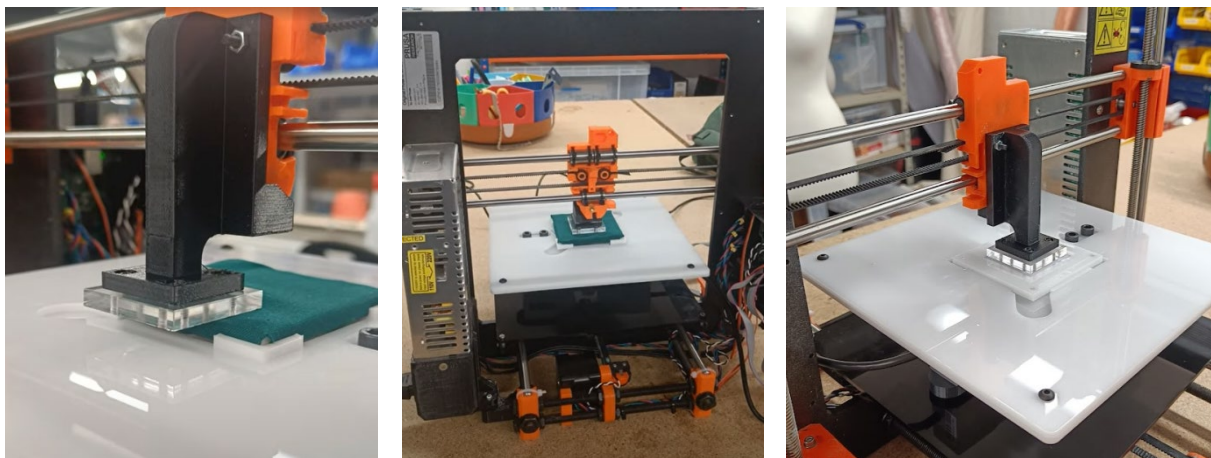


Figure 6 - Details of the adapted PRUSA 3D printer



Figure 7 - Force data collected during (left to right) a) calibration b) calibration c) contact separation testing d) sliding electrode testing

```

triboelectric_test_program.py
C:\Users\sushi> OneDrive\Documents\Imperial> DE3> Becky placement> triboelectric testing - CODE> triboelectric
1194 def calibrate_z(z_coord=85, step=0.1):
1195     """
1196     """
1197     print("Neutral calibration starting")
1198     global latest_force, neutral_z_coord, separation_z_coord, contact_z_coord
1199
1200     neutral_calibration = True
1201
1202     # --- Loop 1: Find neutral z ---
1203     while neutral_calibration:
1204         send_gcode("G1 Z(z_coord) F100") # slow movement
1205         time.sleep(0.1) # give printer time to move and force to update
1206
1207         if latest_force is None:
1208             continue # wait until first force reading arrives
1209
1210         if latest_force < touch_threshold:
1211             z_coord -= step # move down by step
1212         else:
1213             neutral_z_coord = z_coord # sets the z at which the material is just being touched
1214             neutral_calibration = False
1215             print(f"neutral_z_coord = (neutral_z_coord)")
1216             print("Neutral calibration complete")
1217
1218     force_calibration = True
1219
1220     # --- Loop 2: Find contact force z ---
1221     while force_calibration:
1222         send_gcode("G1 Z(z_coord) F100")
1223         time.sleep(0.1)
1224
1225         if latest_force < contact_force:
1226             z_coord -= step # move down by step
1227         else:
1228             contact_z_coord = z_coord # sets the z at which desired force is delivered
1229             force_calibration = False
1230             print(f"contact_z_coord = (contact_z_coord)")
1231             print("Force calibration complete")
1232
1233     separation_z_coord = neutral_z_coord + separation_height
1234     send_gcode("G1 Z(separation_z_coord)")
1235
1236     return neutral_z_coord, contact_z_coord, separation_z_coord
1237
1238 def reset():
1239     speed(fast_speed)
1240     send_gcode("G1 X127 Y107 Z200")
1241
1242

```

Figure 8 - Example code function

1.2.3 – Website rebranding

I updated the outdated website to correct information, better show the full depth of work, and align with Imperial's new brand identity [Figure 9].

Achievements

- › Completed T4 training and independently updated content types across all main and sub-pages.
- › Added new content, including alumni, research themes, outreach resources, and EU funded project work.

- › Created a library of stock photos to refresh the lab's image.

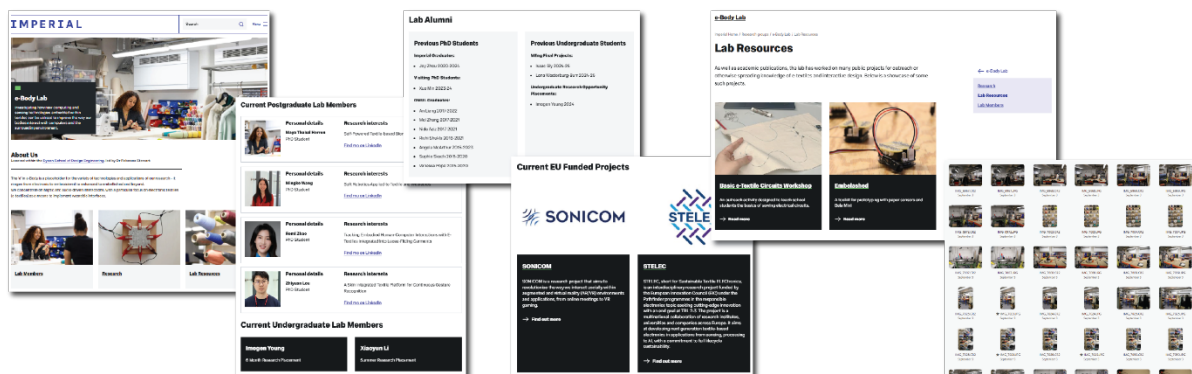


Figure 9 - Selection of updates added to the e-Body website and the new photo library

1.3 – Prior expectations

I expected academia to be focused on research and experiments, but I was surprised by the importance of self-promotion and community engagement. PhD contributed not only to their research question, but also separate papers for EU projects or in collaboration with other universities. With only one full-time academic and a small number of PhDs, I was able to take ownership of outreach and website management—responsibilities I had not anticipated.

2 – PROFESSIONAL DEVELOPMENT

2.1 – Use of Design Engineering knowledge

Strengths:

I applied CAD, 3D design, and coding skills to create testing rigs for Maya's research. I followed design for assembly considerations, making modular designs with consistent M3 fixtures and no adhesives for rapid iterations. When the print head attachment restricted calibration of the x-axis, it was simple to redesign the mount only and keep the original electrode. My computing skills integrated the load cell, Arduino, and 3D printer via G-code, creating a faster testing system

Weaknesses:

As I have no background in materials or triboelectricity, I didn't know what parameters were important when building test protocols, overlooking the importance of electrode separation distance for example. By reading academic papers and asking targeted questions, I improved my understanding to confidently apply my engineering skills to a new application.

2.2 – Application of theoretical and practical methods

Strengths:

I consistently used an iterative design process, getting regular feedback to determine successes and improvements. This was crucial to get all opinions when re-designing the website, as all lab members are represented by work I was completing individually. I additionally used user-centred principles to listen critically and translate Maya's issues into actionable tasks. For instance, identifying what parameters are key to change (e.g. contact time and separation distance) and developed an easy-to-use interface in command line which can change independent variables and run automatically updated test protocols with one command.

Weaknesses:

I found reliance on external feedback frustrating when it caused delays. For example, slow replies from the primary school meant significant changes being made in limited timeframes. Basic information (time allocated and student skill level) given originally was incorrect, meaning parts of the activity needed to be scrapped with less than 48 hours' notice. I adapted to work with unreliable parties, creating a variable plan with each section being able to stand alone if needed. This highlighted the flaws of engineering methods in real-world scenarios, and importance of flexibility.

2.3 – Provision of technical and commercial leadership

Strengths:

I independently led two major projects: the outreach workshop and website redesign. I sourced cost-effective bulk components for the school to buy and tailored the workshop to use equipment they already had. During the session, I took lead teaching and coordinated staff to answer easier questions, leaving me free to solve trickier problems. Despite only a few students completing the

entire activity, both the students and staff learned about e-textiles and showed enthusiasm for more sessions in the future, which were my intended success criteria.

When developing the website, I proactively reached out to T4 administrators to push content live, and regularly presented to the lab, taking on board their feedback and updating as appropriate. I received feedback from my supervisor that "... this is really great work. The lab is represented so much better now" and was thanked for taking control of this task.

Weaknesses:

As my projects were mainly individual, I initially overestimated what I could produce, overcommitting as if I was supported by a team. I learned to understand my limits and accurately predict time needed for tasks. This meant at times rejecting work, such as saying working on an additional paper, as I knew I was at capacity with Maya's project. I now recognise leadership means knowing when to say no to ensure high-quality work.

2.4 – Effective interpersonal skills

Strengths:

I showed excellent communication across diverse audiences, from children with no engineering background, to highly knowledgeable academics. When working with Maya, I updated her on my progress via teams or in person meetings. I also showed good communication skills through visual resources, making worksheets accessible for children through images and plain written text.

Weaknesses:

I often waited for direct prompts to share updates, rather than being proactive, often leading me to spend longer than needed trying to solve an issue. To improve during my masters, I have created a template for weekly updates and will be reminding myself with weekly calendar alerts. I also recognised my reluctance to share negative information, such as getting stuck or making a mistake. I learned honest communication with managers prevents issues escalating and builds trust.

2.5 – Personal commitment to professional standards

Strengths:

I followed all codes of conduct and safety procedures in the lab, as well as safeguarding policies when interacting with school students. Regular meetings with Becky held me accountable and helped manage workload responsibly.

Weaknesses:

I struggled with insomnia during part of my placement, missing an important meeting and initially avoiding addressing the issue. This was a clear lapse in my professionalism. However, I later held myself accountable and discussed openly with Becky. I learned to share when you have issues in your personal life, and I only made meeting commitments in the afternoon for a period to ensure I could attend. Being professional doesn't mean ignoring your own needs, just making sure to communicate them clearly with others.

3 – FUTURE CAREER REFLECTION

3.1 – Personal growth

This placement not only developed my engineering skills, but also my understanding of soft skills in the work environment. I had previously believed showing mistakes was a weakness and that a 'model employee' should only ever showcase success. I learned through experience that being honest even when you make mistakes is much more important. Whilst this was difficult to confront face on, I learned how important transparency and accountability are. This experience changed my approach to professional conduct and taught me that what matters most is how you respond to and grow from those situations.

I also discovered how I work best. Flexible working hours accommodate the highs and lows of productivity from ADHD, as I have come to accept my workflow won't always be consistent like others. I thrived when given discrete individual tasks to complete at my own pace, while working towards a collaborative goal and having a team around to work and socialise with.

3.2 Professional growth

Professionally, I advanced my ability to apply rapid prototyping skills in a research context, and to areas I have no prior technical knowledge of, producing test rigs and equipment to directly support PhD-level work. I also learned how academic publications are written, through observing lab practice and contributing as co-author to an upcoming CHI submission. Exposure to this process gave me confidence to transition from university level testing to formal academic experimentation.

Beyond technical knowledge, I developed confidence in outreach and teaching. Designing worksheets, running workshops, and presenting activities at public events showed me how much I love making engineering accessible and inspiring others to enjoy STEM. This highlighted the potential desire to combine research and public engagement in my future career.

3.3 – Strengths, weaknesses, and discoveries

Strengths:

I demonstrated creativity and adaptability, thinking outside the box to solve problems quickly. I also showed a clear skill in independent work with limited direction, using a broad spectrum of hardware and software skills across new fields of engineering.

Weaknesses:

I learned I need to become more active in communication and not avoid challenging conversations. During my masters, I plan to implement structured updates and make more deliberate use of peer feedback to avoid working in isolation. I also will continue working on recognising my limits and communicating my capacity when working in groups or reporting to managers.

Discoveries:

Flexible working hours are vital for me to work effectively, especially in accepting how ADHD impacts my productivity. I also discovered how much I enjoy teaching and outreach, which I had not previously considered a central part of my career path.

3.4 – Further industry opportunities

This placement confirmed my desire to pursue a career in research and academia, starting by completing a PhD. I value the freedom and flexibility research provides, alongside the sense of community from collaboration. I particularly enjoy projects with combine practical prototyping with material development and want to take this further to build new user centred interactions from e-textile innovations.

To prepare for my goal of a PhD, I have begun taking practical steps, discussing with Becky and Maya how the application process works, and where funding typically comes from. I aim to continue working on triboelectricity through a master's with Becky, researching applications of triboelectric sensing to build a novel interaction. I will also be working towards a publication alongside Maya to characterise the currently unknown triboelectric properties of biomaterials. I also now recognise that academia comes hand in hand with teaching and hope to continue working as a UTA to prepare me for working within universities. I am additionally seeking out external volunteering roles in science communication, such as mentoring secondary school students in a robotics competition via RoboCon [4].

As I enter my final undergraduate year, I see this placement as a springboard towards an e-textiles PhD, potentially within the e-Body Lab itself, and as the foundation of a long-term academic career.

References:

1. Times Higher Education. REF 2021: Outputs scores [Internet]. Times Higher Education (THE). 2022 [cited 2025 Sep]. Available from: <https://www.timeshighereducation.com/news/ref-2021-outputs-scores>
2. e-Body Lab | Research groups | Imperial College London [Internet]. Imperial.ac.uk. 2025 [cited 2025 Sep]. Available from: <https://www.imperial.ac.uk/e-body-lab/>
3. Sewing Simple Circuits [Internet]. Imperial College London. 2025 [cited 2025 Sep]. Available from: <https://www.imperial.ac.uk/e-body-lab/lab-resources/sewing-simple-circuits/>
4. RoboCon [Internet]. Robocon.uk. 2025 [cited 2025 Sep]. Available from: <https://robocon.uk/>